SALES DATA ANALYSIS

Internship Project Report | Python, Pandas, NumPy and Data Visualization

Project objective: Analyze sales transactions to understand revenue trends, regional performance, product-category performance, and customer segments. Demonstrate a repeatable data-cleaning and exploratory-analysis workflow.

Dataset note: The included dataset is synthetic and illustrative, created for demonstrating the analysis workflow. It is not actual company data. Replace it with the official dataset before submitting real business conclusions.

1. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn and CSV data files.

2. Methodology

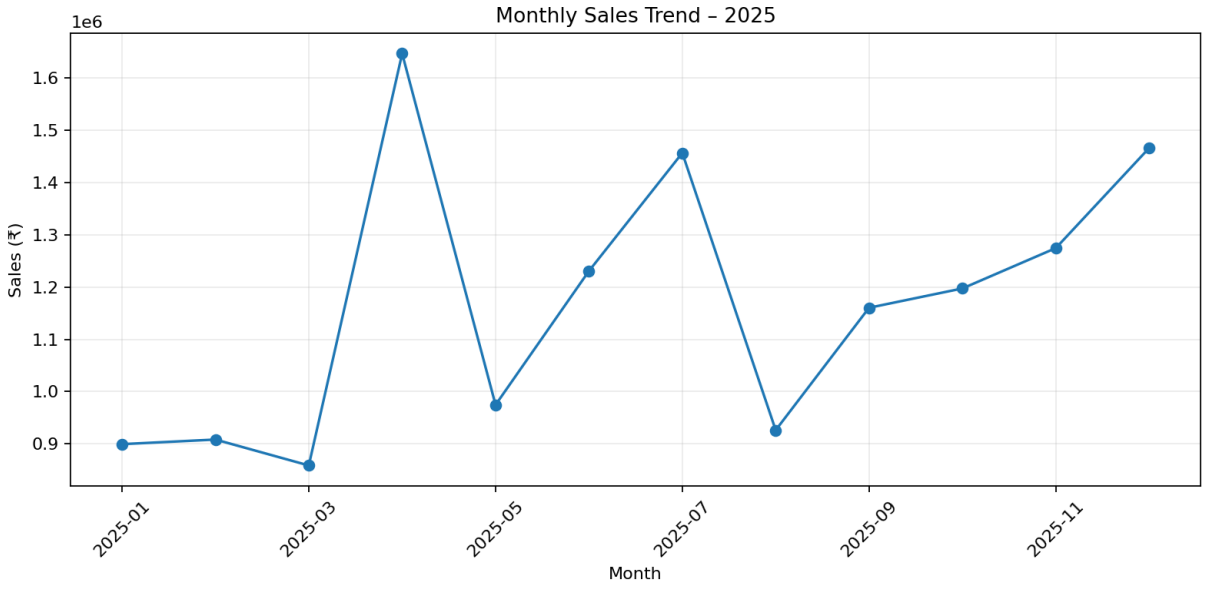
The workflow includes loading the dataset, inspecting data types and missing values, removing duplicate rows, filling missing region values with the mode, validating dates, calculating net sales after discounts, grouping results, and preparing visualizations.

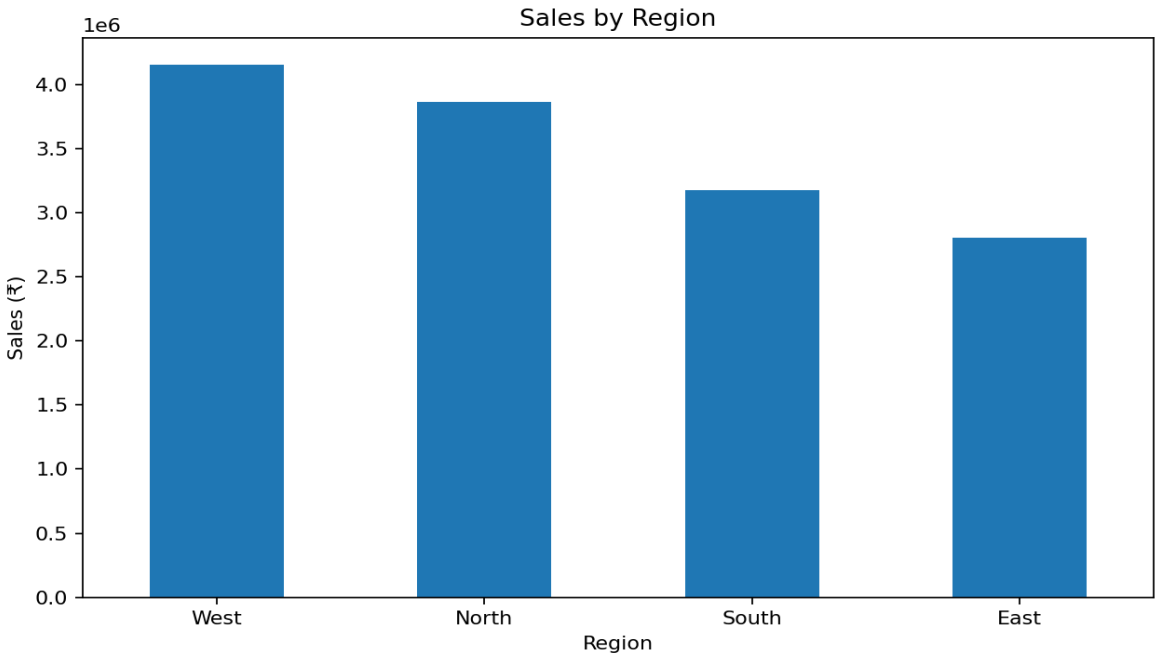
3. Key Performance Indicators

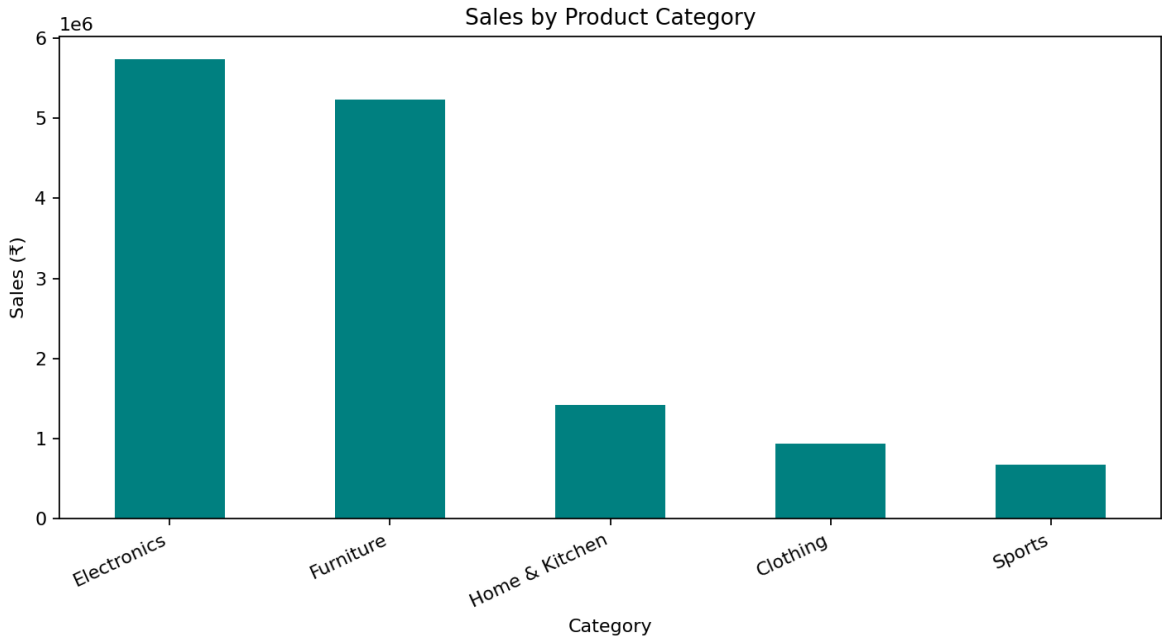
Metric	Result
Total sales	■13,999,724.74
Unique orders	600
Average sales per order	■23,332.87
Top region by sales	West
Top category by sales	Electronics
Top product by sales	Office Chair

4. Visual Analysis

The charts below show monthly revenue movement and sales distribution across regions and categories.




Sales by Region
1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Sales (₹)
West
North
South
East
Region



5. Findings and Business Insights

• The highest-sales region in this sample is **West**.
• The highest-sales category is **Electronics**.
• The highest-sales product is **Office Chair**.
• Monthly analysis can help identify seasonal demand and plan inventory.
• Customer-type analysis can support retention and acquisition decisions.

6. Recommendations

1. Maintain sufficient stock for consistently high-performing products.
2. Investigate low-performing regions and categories using customer feedback and local demand.
3. Use monthly trends to plan promotions and inventory.
4. Track repeat-customer sales and evaluate retention campaigns.
5. Refresh this report regularly using verified business data.

7. Conclusion

This project demonstrates an end-to-end sales data analysis process, from data preparation to visual reporting. The approach can be reused with an organization's actual sales data to support evidence-based decisions.

Prepared as an educational internship project. Replace synthetic data and sample findings with verified organization data where required.